

SEVERE WEATHER ENHANCERS

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Three characteristics that determine how severe a warm season thunderstorm will be are [low level speed and directional wind shear](#), the [upper level wind](#), and [instability](#) (amount of positive buoyancy). Strong surface wind adds fuel to a storm just as an influx of oxygen adds fuel to a fire. Strong surface winds transport [moisture](#) and momentum into a developing storm. Terms such as [moisture convergence](#) and surface inflow describe this process. Directional shear causes low level wind rotations which can cause [tornadic](#) conditions to be more favorable. The upper level wind has many effects on a thunderstorm, which include: a sheared anvil, and upper level divergence (mass is removed from the top of the storm which intensifies the upward pressure gradient within a storm). Upper level winds (between 500 and 300 mb level) of greater than 100 knots is considered significant. The amount of buoyancy determines the updraft strength. The warmer the air within the storm is (compared to the surrounding environmental air), the faster the air will rise within the storm. [CAPE](#) values of greater than 1,500 J/kg and LI values less than -5 indicates a very buoyant troposphere. All these factors and others combine together to determine the [severity](#) of a thunderstorm.

A trigger mechanism is any process that initiates precipitation or storm development. It is in reference to a process that causes a precipitation or storm event and without this process precipitation or storms would not have occurred. Common trigger mechanism examples are [lifting mechanisms](#), increase of low level moisture, daytime heating, instability and wind shear. The most common type of trigger mechanism that will be referenced are lifting mechanisms such as fronts and other low level [convergence](#) boundaries. Severe weather situations will often have several triggers. The trigger that initially sets the storms into motion will often be referenced as the trigger for the severe storm event.

FORECASTING SEVERE WEATHER USING SKEW-T

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One of the most important times to examine soundings is during times when severe weather is likely. Skew-T's can give you a general idea of the character of the severe weather. Below are severe weather phenomena and how to generally identify it's potential from the Skew-T diagram.

Strong straight-line WINDS:

Look for a hydrolapse and large dewpoint depressions in the mid-levels of the troposphere. Winds will also occur in association with an inverted-V sounding. The moist air parcels from the storm mixes with the surrounding dry air. This evaporational cooling produces negative buoyancy, causing air to accelerate toward the surface. High based storms will generally have stronger winds since the downdrafts have a longer distance above the surface to accelerate to the surface.

LARGE HAIL:

Lower values of PW (precipitable water) preferred. Large PW values will water load the updraft. For large hail you need a large updraft and thus large CAPE; High PW impedes this. PW less than 1.25 inches is relatively low. PW above 1.75 will significantly water load the updraft. LP and classic supercells have largest hail. Large PW (i.e. greater than 2.0 inches, can reduce upward vertical velocity of updraft by more than half).

As mentioned, the more CAPE the better. Hail is more likely in high elevation areas since the freezing level is closer to the surface. A low freezing level is beneficial for hail since the hailstones will not have as much time to melt before they hit the ground. A supercell is needed to produce large hail. Look for loaded gun sounding and convective instability.

Click [here](#) for an in-depth explanation of hail formation.

TORNADO:

Strong veering of wind in boundary layer. Look for loaded gun sounding with plenty of convective instability. Strong upper level jet will tilt thunderstorm, ensuring it will be a supercell. MUST have winds in the boundary layer averaging above 20 knots. Strong low level jet along with veering boundary profile adds large storm relative inflow into storm. This produces large Helicity values. There needs to be a good balance between shear and instability.

Click [here](#) for an in-depth explanation of tornado formation

HEAVY RAIN (flash flood):

High PW value, well above climatological norm. Strong low level forcing but with relatively weak upper level wind. Moisture convergence into stationary low level feature (such as a stationary front, tropical circulation).